PGGAN Based Tumor Progression Detection

Harshdip Saha
Netaji Subhas University of Technology
Dwarka Sector-3, Delhi, India
Email: harshdipsaha@gmail.com

***Abstract*—Augmenting paired pre- and post-treatment *Magnetic Resonance Imaging (MRI)* scans is essential for robust modeling of tumor progression, yet most existing generative approaches focus on synthesizing isolated time points rather than capturing temporal relationships between scans. In this work, we propose a novel dual-generator, dual-discriminator framework based on the *Progressive Growing of Generative Adversarial Networks (PGGAN)* paradigm, explicitly designed to generate anatomically consistent pre- (Referring as Baseline Scan) and post- (Referring as Follow-up Scan) treatment 3D MRI volumes for the same patient. The proposed architecture employs two coordinated generators—one to produce baseline scans from noise, and a second that generates follow-up scans conditioned on the synthetic baseline through cross-stage feature merging. We introduce L1 and *Structural Similarity Index Measure (SSIM)*–based temporal consistency losses to bridge the gap between paired time points, ensuring that the generated follow-up images maintain realistic structural changes relative to their baselines. This approach facilitates the creation of large, synthetic paired datasets that preserve critical temporal dynamics, enabling more realistic data augmentation for longitudinal tumor studies and supporting improved downstream analysis. Experimental results demonstrate enhanced anatomical consistency and visual fidelity in generated scan pairs, laying the groundwork for robust data augmentation and methodological advances in progression-aware medical image synthesis.**

***Index Terms*—Generative Adversarial Networks, PGGAN, Data Augmentation, Medical Imaging, MRI, Brain Tumors, Longitudinal Analysis.**

I. Introduction

Longitudinal analysis of brain tumor progression relies on the availability of paired pre- and post-treatment MRI scans for each patient. There are four categories in which response to a treatment is divided, based on the **Response Assessment in Neuro-Oncology (RANO)** criteria [1]: Class 0: Complete response, Class 1: Partial response, Class 2: Stable Disease, Class 3: Progressive Disease. However, generating synthetic paired samples remains an unsolved challenge in medical image augmentation, largely due to the complex temporal relationship between scans and severe dataset limitations.

Generative Adversarial Networks (GANs) introduced the adversarial training paradigm for realistic sample synthesis and representation learning [2]. Prior GAN-based augmentation works have demonstrated the value of synthetic imaging for single timepoints or modality translation [3], [4], but most focus on individual timepoints rather than paired longitudinal generation [5], [6].

The ability to simulate paired pre- and post-treatment imaging would be transformative for clinical research and machine learning in neuro-oncology. It would enable advanced automated tumor progression detection systems, reduce the reliance on manual expert annotation, and support robust training of segmentation and classification models even in the face of limited real paired longitudinal data.

Our work was inspired by the recently introduced **Brain Tumor Segmentation (BraTS)** Lighthouse Challenge and its progression subchallenge [7], [8]. The released dataset, although still limited in size, highlights the real-world barrier in longitudinal sample acquisition and the urgent need for effective paired augmentation strategies.

We address this gap by proposing a dual-generator, dual-discriminator PGGAN architecture for realistic augmentation of paired pre- and post-treatment 3D MRI volumes. Our approach leverages cross-stage feature merging and novel temporal loss functions to explicitly enforce anatomical and structural consistency between paired scans. To our knowledge, this is among the first attempts to generate longitudinally paired synthetic brain MRI samples—providing a new foundation for progression-aware model training, methodological innovation, and potential clinical adoption in automated tumor growth analysis.

II. Related Work

Generative Adversarial Networks (GANs) have become a cornerstone in medical image analysis due to their capacity to generate realistic synthetic data, address data scarcity, and advance applications in reconstruction, segmentation, and disease progression modeling [4], [9]–[11]. Early GAN-based approaches primarily focused on image-to-image translation, augmentation, and super-resolution [5], [6], [12], offering improvements in single timepoint medical image synthesis and downstream tasks such as classification and segmentation [3].

The transition to 3D and volumetric image synthesis brought further advancements [13]. Networks such as MedGAN and **Variational Autoencoder (VAE)**-GAN hybrids [14] have enabled generation of full volumetric MR images, motion correction, and robust cross-domain translation between modalities [10], [15]. These approaches support diverse tasks, including domain adaptation, denoising, and motion artifact correction. Despite these advances, most GAN architectures for medical imaging have focused on generating either baseline or follow-up images separately, or on unidirectional domain shifts (e.g.,

healthy to diseased), rather than synthesizing temporally linked pairs.

A persistent limitation is the lack of models that jointly learn the temporal relationship between paired scans for longitudinal analysis [11]. While conditional GANs and paired translation models such as Pix2Pix [6], MUNIT [16], and SimGAN [17] have enabled some limited applications in paired or conditional generation, they often require a real baseline as input or, as in the work by Han et al. [18], are used as a two-step 2D refinement process rather than a direct 3D longitudinal synthesis. These models do not jointly produce both pre- and post-treatment 3D volumes in a fully generative setup.

Recent reviews highlight the demand for new architectures capable of addressing longitudinal modeling, data imbalance, and realistic paired data generation for progression-aware medical AI [9], [11]. Our work responds to these challenges by introducing a dual-generator, dual-discriminator framework based on Progressive Growing of GANs (PGGAN) [19], specifically designed to generate anatomically consistent paired MR images and capture clinically relevant temporal progression dynamics.

III. Materials and Preprocessing

A. Dataset

In this work, we use the longitudinal multi-class brain tumor MRI dataset released as part of the BraTS Lighthouse Challenge 2025 [7]. The dataset includes paired pre-treatment (baseline) and post-treatment (follow-up) 3D MRI scans for four tumor progression categories (Class 0: 27 patients, Class 1: 21 patients, Class 2: 86 patients, Class 3: 231 patients). Images originate from multiple institutions and exhibit significant anatomical and acquisition variability, reflecting real-world heterogeneity and limited sample size for paired longitudinal data.

Each patient folder represents one subject and consists of ten 3D **Neuroimaging Informatics Technology Initiative (NIfTI)** (`.nii.gz`) volumes. These ten volumes correspond to multiple MRI modalities at two timepoints (baseline and follow-up), along with their respective segmentation masks. Specifically, each patient folder contains the following files:

- Four baseline MRI modalities: **T1-weighted (T1)**, **Contrast-enhanced T1 (T1ce)**, **T2-weighted (T2)**, and **Fluid-Attenuated Inversion Recovery (FLAIR)**
- Four follow-up MRI modalities: T1, T1ce, T2, and FLAIR
- One baseline segmentation mask
- One follow-up segmentation mask

B. Preprocessing Pipeline

A robust preprocessing pipeline was implemented to prepare the paired MRI scans for GAN-based learning. The pipeline steps are:

1) **Canonical Orientation:** All MRI volumes were first converted to the canonical **RAS+ (Right, Anterior, Superior)** coordinate orientation using TorchIO's `ToCanonical()` transform to ensure spatial consistency across all subjects and modalities [20]. The RAS+ convention defines voxel axes to increase toward the Right (R), Anterior (A), and Superior (S) directions, thereby standardizing anatomical alignment across heterogeneous acquisition protocols. Consistent axis orientation is crucial for volumetric registration, patch sampling, and convolutional operations in 3D GAN architectures, preventing spatial ambiguity during training.
2) **Registration:** Each follow-up scan was rigidly registered (6 degrees of freedom) to its corresponding baseline T1-weighted volume using mutual information as the similarity metric. Rigid registration preserves intracranial structure and tumor morphology while aligning scans in a common anatomical space, enabling the model to focus on longitudinal appearance changes rather than inter-subject variations.
3) **Resampling and Cropping:** All co-registered volumes were then resampled to an isotropic voxel size of $1\,\mathrm{mm}^3$ and cropped or padded to a uniform spatial dimension of $128 \times 128 \times 128$ voxels. Isotropic resampling ensures equal physical spacing along all axes, which stabilizes convolutional receptive fields. Uniform volume size simplifies GPU batching and enables progressive 3D generation in the proposed TPGGAN model. (Volumes are stored in NIfTI format [21].)
4) **Brain Extraction:** A brain extraction step was applied using an Otsu-based threshold mask on the baseline T1 weighted image, followed by masking of other modalities [22]. Skull and non-brain tissues introduce high-intensity outliers that can bias GAN training. While deep-learning-based skull-stripping methods (e.g., **HD-BET (High-Definition Brain Extraction Tool)** [23]) can achieve higher precision, the Otsu-based approach was adopted here for computational efficiency and full automation.

$$t^* = \arg\min_t \sigma^2_{\text{within}}(t) \tag{1}$$

$$\text{mask}(x) = \begin{cases} 1, & x > t^* \\ 0, & x \leq t^* \end{cases} \tag{2}$$

5) **Intensity Normalization:** After masking, each modality was z-score normalized within the brain region using per-subject mean and standard deviation:

$$x_i' = \frac{x_i - \mu_{\text{brain}}}{\sigma_{\text{brain}}} \tag{3}$$

where μ_{brain} and σ_{brain} represent the mean and standard deviation of brain-region voxels. MRI intensity scales are non-standardized across scanners; per-volume normalization minimizes inter-subject variability and improves model convergence stability.
6) **Quality Control:** Finally, automated **Quality Control (QC)** was performed by saving montage visualizations

Fig. 1: Preprocessing for Paired MRI GAN Training

for each patient in dataset **do**
  Load all modalities and segmentations:
    $subject \leftarrow \text{TorchIOSubject}(t1_base, \ldots, mask_foll$
  $subject \leftarrow \text{ToCanonical}(subject)$
  for each volume in subject **do**
    $volume \leftarrow \text{RegisterToReference}(volume, t1_base)$
    $volume \leftarrow \text{ResampleTo1mm}(volume)$
    $volume \leftarrow \text{CropOrPadToShape}(volume, mask = brain_mask, target = (128, 128, 128))$
  end for
  $brain_mask \leftarrow \text{MakeBrainMask}(subject.t1_base)$ ▷ Otsu thresholding
  $subject \leftarrow \text{ZNormalize}(subject, mask = brain_mask)$
  for each volume in subject **do**
    $\text{SaveMontage}(volume, out_dir)$ ▷ axial, sagittal, coronal slice visualization
  end for
  $\text{np.save}(out_dir, subject)$
end for

TABLE I: Content of a patient file after preprocessing

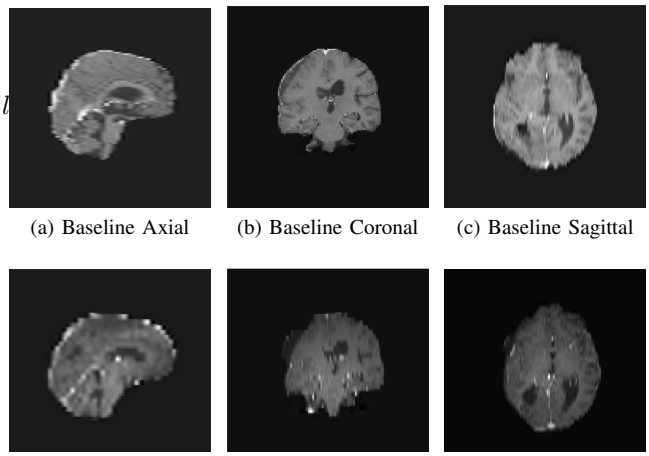
Index	Content	Scan Timepoint
0	T1-weighted (T1)	Baseline
1	Contrast-enhanced T1 (T1ce)	Baseline
2	T2-weighted (T2)	Baseline
3	Fluid-Attenuated Inversion Recovery (FLAIR)	Baseline
4	T1-weighted (T1)	Follow-up
5	Contrast-enhanced T1 (T1ce)	Follow-up
6	T2-weighted (T2)	Follow-up
7	Fluid-Attenuated Inversion Recovery (FLAIR)	Follow-up
8	Segmentation mask	Baseline
9	Segmentation mask	Follow-up

of central axial, sagittal, and coronal slices for each processed subject. Visual QC enables rapid verification of registration accuracy, skull-stripping success, and normalization consistency prior to GAN training.

The final shape of each patient is [10,128,128,128] and stored as .npz files which is used for unified data loading and model input during training.

IV. Model Architecture

In this study, we introduce the **Temporal Progressive Growing GAN (TPGGAN)**, a dual-stage adversarial framework designed to synthesize anatomically consistent longitudinal MRI volumes representing pre-treatment and post-treatment tumor states. Unlike conventional GAN methods that generate single time-point images, TPGGAN explicitly models tumor evolution by generating a baseline brain MRI volume and a corresponding follow-up scan conditioned on both latent structure and inferred baseline anatomy.



(a) Baseline Axial (b) Baseline Coronal (c) Baseline Sagittal

(d) Follow-up Axial (e) Follow-up Coronal (f) Follow-up Sagittal

Fig. 2: Original paired data from BraTS 2025, showing central slices for Baseline (Top) and Follow-up (Bottom) scans in Axial, Coronal, and Sagittal views.

A. Overview

TPGGAN comprises two generators (G_{base} and G_{follow}) and two independent discriminators (D_{base} and D_{follow}). Both generators follow the progressive growing paradigm, beginning at low resolution (4^3) and incrementally expanding to the full (128^3) voxel resolution. At each growth stage, new convolutional blocks are faded in smoothly, stabilizing training and preserving coarse-to-fine anatomical detail [19].

- **Baseline Generator $G_{base}(z)$:** Synthesizes a pre-treatment MRI volume from a random latent vector $z \sim \mathcal{N}(0, I)$. Its architecture is detailed in Table II.
- **Follow-up Generator $G_{follow}(z', G_{base}(z))$:** Produces the post-treatment MRI volume, conditioned jointly on a second latent code z' and the baseline anatomical features. Its architecture is detailed in Table III.
- **Dual Discriminators:** D_{base} and D_{follow} independently enforce realism in baseline and follow-up domains, preventing collapse and ensuring structure is preserved across time. Their shared architecture is detailed in Table IV.

B. Progressive Growing Backbone

Both generators and discriminators adopt the Progressive Growing of GANs (PGGAN) strategy, inspired by [19]. Resolution doubles stage-by-stage, with each step adding:

$$\text{Conv3D} \rightarrow \text{LeakyReLU} \rightarrow \text{Upsample} \tag{4}$$

for generators, and the inverse for discriminators. A fade-in parameter $\eta \in [0, 1]$ blends old and new layers to prevent convergence instability.

All convolutional layers employ weight scaling, pixelwise feature normalization, and equalized learning rate initialization to improve signal propagation and stability in 3D.

TABLE II: TPGGAN Generator (G_{base}) Architecture. All layers are 3D. LReLU denotes LeakyReLU (0.2). All blocks include PixelNorm.

Layer	**Activation**	**Output Shape (Channels × D × H × W)**
Latent vector z	-	$256 \times 1 \times 1 \times 1$
ConvTranspose $4 \times 4 \times 4$	LReLU	$256 \times 4 \times 4 \times 4$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 4 \times 4 \times 4$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 4 \times 4 \times 4$
Upsample (Trilinear)	-	$256 \times 8 \times 8 \times 8$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 8 \times 8 \times 8$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 8 \times 8 \times 8$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 8 \times 8 \times 8$
Upsample (Trilinear)	-	$256 \times 16 \times 16 \times 16$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 16 \times 16 \times 16$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 16 \times 16 \times 16$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 16 \times 16 \times 16$
Upsample (Trilinear)	-	$256 \times 32 \times 32 \times 32$
WSConv $3 \times 3 \times 3$	LReLU	$128 \times 32 \times 32 \times 32$
WSConv $3 \times 3 \times 3$	LReLU	$128 \times 32 \times 32 \times 32$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 32 \times 32 \times 32$
Upsample (Trilinear)	-	$128 \times 64 \times 64 \times 64$
WSConv $3 \times 3 \times 3$	LReLU	$64 \times 64 \times 64 \times 64$
WSConv $3 \times 3 \times 3$	LReLU	$64 \times 64 \times 64 \times 64$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 64 \times 64 \times 64$
Upsample (Trilinear)	-	$64 \times 128 \times 128 \times 128$
WSConv $3 \times 3 \times 3$	LReLU	$32 \times 128 \times 128 \times 128$
WSConv $3 \times 3 \times 3$	LReLU	$32 \times 128 \times 128 \times 128$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 128 \times 128 \times 128$

TABLE III: TPGGAN Conditional Generator (G_{follow}) Architecture. At each stage, features are merged with the downsampled baseline scan.

Layer	**Activation**	**Output Shape (Channels × D × H × W)**
Latent vector z'	-	$256 \times 1 \times 1 \times 1$
ConvTranspose $4 \times 4 \times 4$	LReLU	$256 \times 4 \times 4 \times 4$
Concat (with downsampled Baseline $1 \times 4 \times 4 \times 4$)	-	$257 \times 4 \times 4 \times 4$
Merge (WSConv $3 \times 3 \times 3 \times 2$)	LReLU	$256 \times 4 \times 4 \times 4$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 4 \times 4 \times 4$
Upsample (Trilinear)	-	$256 \times 8 \times 8 \times 8$
WSConv $3 \times 3 \times 3 \times 2$ (G_Block)	LReLU	$256 \times 8 \times 8 \times 8$
Concat (with downsampled Baseline $1 \times 8 \times 8 \times 8$)	-	$257 \times 8 \times 8 \times 8$
Merge (WSConv $3 \times 3 \times 3 \times 2$)	LReLU	$256 \times 8 \times 8 \times 8$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 8 \times 8 \times 8$
... (Blocks repeat for 16^3, 32^3, 64^3) ...	...	...
Upsample (Trilinear)	-	$64 \times 128 \times 128 \times 128$
WSConv $3 \times 3 \times 3 \times 2$ (G_Block)	LReLU	$32 \times 128 \times 128 \times 128$
Concat (with Baseline $1 \times 128 \times 128 \times 128$)	-	$33 \times 128 \times 128 \times 128$
Merge (WSConv $3 \times 3 \times 3 \times 2$)	LReLU	$32 \times 128 \times 128 \times 128$
to_rgb (WSConv $1 \times 1 \times 1$)	*Linear*	$1 \times 128 \times 128 \times 128$

C. Conditional Tumor-Progression Modeling

To model tumor evolution, follow-up generation is conditioned on both the baseline image and latent perturbation:

$$G_{follow}(z', G_{base}(z)) = F\big(G_{base}(z), z'\big) \tag{5}$$

As detailed in Table III, the architecture of G_{follow} is not a simple generator. At each resolution stage (from 4^3 to 128^3), the synthetic baseline scan ($G_{base}(z)$) is downsampled to the current resolution and concatenated (as a 1-channel volume) to the generator's feature maps. This combined feature map

TABLE IV: TPGGAN Discriminator (D_{base}/D_{follow}) Architecture. All layers are 3D. LReLU denotes LeakyReLU (0.2).

Layer	**Activation**	**Output Shape (Channels × D × H × W)**
Input Image	-	$1 \times 128 \times 128 \times 128$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$32 \times 128 \times 128 \times 128$
WSConv $3 \times 3 \times 3$	LReLU	$32 \times 128 \times 128 \times 128$
WSConv $3 \times 3 \times 3$, AvgPool3d	LReLU	$64 \times 64 \times 64 \times 64$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$64 \times 64 \times 64 \times 64$
WSConv $3 \times 3 \times 3$	LReLU	$64 \times 64 \times 64 \times 64$
WSConv $3 \times 3 \times 3$, AvgPool3d	LReLU	$128 \times 32 \times 32 \times 32$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$128 \times 32 \times 32 \times 32$
WSConv $3 \times 3 \times 3$	LReLU	$128 \times 32 \times 32 \times 32$
WSConv $3 \times 3 \times 3$, AvgPool3d	LReLU	$256 \times 16 \times 16 \times 16$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$256 \times 16 \times 16 \times 16$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 16 \times 16 \times 16$
WSConv $3 \times 3 \times 3$, AvgPool3d	LReLU	$256 \times 8 \times 8 \times 8$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$256 \times 8 \times 8 \times 8$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 8 \times 8 \times 8$
WSConv $3 \times 3 \times 3$, AvgPool3d	LReLU	$256 \times 4 \times 4 \times 4$
from_rgb (WSConv $1 \times 1 \times 1$)	*LReLU*	$256 \times 4 \times 4 \times 4$
MinibatchStdDev	-	$257 \times 4 \times 4 \times 4$
WSConv $3 \times 3 \times 3$	LReLU	$256 \times 4 \times 4 \times 4$
Flatten	-	16384
Fully-connected ($16384 \rightarrow 256$)	LReLU	256
Fully-connected ($256 \rightarrow 1$)	Linear	1

is then passed through a dedicated **merge block** (two 3D convolutional layers) before proceeding to the next upsampling stage. This mechanism, also visualized in Fig. 3, forces the follow-up generator to be aware of the baseline anatomy at every spatial scale.

Feature interaction occurs through 3D convolutional fusion blocks at each scale, allowing:

1) global structure to propagate from early layers,
2) local tumor-specific deformation to emerge with resolution growth,
3) progressive anatomical alignment between baseline and follow-up volumes.

This mechanism enforces clinically realistic tumor growth instead of arbitrary structural drift.

D. Discriminator Design

Each discriminator (Table IV) mirrors its generator counterpart through a progressive downsampling pathway, employing 3D convolutional fusion blocks to aggregate multi-scale spatial features. To improve stability and mitigate mode collapse, a minibatch-standard-deviation regularization term is incorporated as:

$$D(x) = \sigma(f(x) + \mathrm{mbstd}(x)), \tag{6}$$

where $f(x)$ denotes the learned 3D feature representation and $\mathrm{mbstd}(x)$ captures the cross-sample variance within the minibatch. This variance term is appended as an additional feature channel before the final layer, enabling the discriminator to detect lack of diversity in the generator's outputs.

Wasserstein GAN with Gradient Penalty (WGAN-GP) regularization further stabilizes adversarial training [24], [25].

E. Summary

TPGGAN introduces a clinically guided generative formulation: baseline anatomy and progression dynamics are jointly modeled through a dual-generator, dual-discriminator, progressively grown 3D architecture. This design enables anatomically consistent MRI time-series synthesis where pathological evolution between stages is smooth, plausible, and patient-specific.

V. Optimization and Progressive Training Strategy

The proposed TPGGAN is trained under a Wasserstein adversarial framework with gradient penalty (WGAN-GP), complemented by longitudinal anatomical consistency constraints. Let G_b denote the baseline generator, G_f the follow-up generator, and $\mathcal{D}_b$, $\mathcal{D}_f$ their corresponding discriminators. Two latent vectors $z_b, z_f \sim \mathcal{N}(0, I)$ generate baseline and follow-up volumes

$$\hat{x}_b = G_b(z_b), \qquad \hat{x}_f = G_f(z_f, \hat{x}_b), \tag{7}$$

with G_f conditioned on G_b to enforce longitudinal dependency.

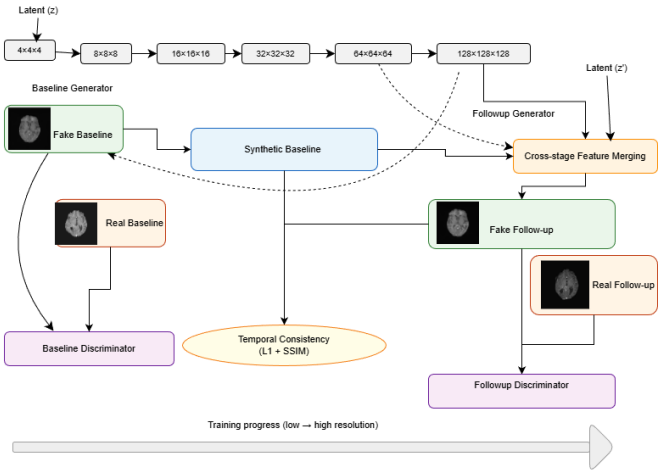


Fig. 3: TPGGAN architecture: baseline MRI is synthesized from z and used, with z', to condition follow-up synthesis.

A. Wasserstein Adversarial Objective

For each timepoint $t \in \{b, f\}$ (baseline and follow-up), the discriminator $\mathcal{D}_t$ serves as a *critic* that estimates the Wasserstein distance between the true data distribution p_{real} and the generator's distribution p_{gen} [24], [25]. This formulation provides a smoother optimization landscape compared to the original Jensen–Shannon–based GAN loss, yielding more stable training for high-dimensional volumetric data.

The critic (discriminator) objective is defined as:

$$\mathcal{L}_{D_t} = E_{x_t}[\mathcal{D}_t(x_t)] - E_{\hat{x}_t}[\mathcal{D}_t(\hat{x}_t)] + \lambda_{gp} E_{\tilde{x}_t}\left[(\|\nabla_{\tilde{x}_t}\mathcal{D}_t(\tilde{x}_t)\|_2 - 1)^2\right], \tag{8}$$

where $\hat{x}_t$ denotes generated samples, and $\tilde{x}_t = \alpha x_t + (1-\alpha)\hat{x}_t$, with $\alpha \sim U(0,1)$. Here, λ_{gp} is the gradient penalty coefficient, which we set to 10 as recommended in [25].

The corresponding generator objective is:

$$\mathcal{L}_{G_t} = -E_{\hat{x}_t}[\mathcal{D}_t(\hat{x}_t)], \tag{9}$$

which encourages the generator to produce synthetic MRI volumes that maximize the critic's response.

B. Longitudinal Consistency Constraints

Since tumor progression exhibits continuous spatial evolution, the generated follow-up volume must remain structurally consistent with the baseline except for disease-driven changes. We enforce anatomical correspondence through L1 and 3D structural similarity (SSIM) constraints [26]:

$$\mathcal{L}_{L1} = \|\hat{x}_f - \hat{x}_b\|_1, \tag{10}$$

$$\mathcal{L}_{SSIM} = 1 - SSIM_{3D}(\hat{x}_f, \hat{x}_b). \tag{11}$$

The follow-up generator objective becomes:

$$\mathcal{L}_{G_f}^{total} = \mathcal{L}_{G_f} + \lambda_1 \mathcal{L}_{L1} + \lambda_{ssim}\mathcal{L}_{SSIM}. \tag{12}$$

Here, λ_1 and λ_{ssim} are weighting parameters to balance the consistency losses with the adversarial loss. In our experiments, we set $\lambda_1 = 10$ and $\lambda_{ssim} = 1$.

C. Progressive Growing and EMA

Both generators and discriminators are trained with progressive resolution expansion:

$$4^3 \to 8^3 \to \cdots \to 128^3, \tag{13}$$

with a linear fade-in coefficient $\eta \in [0, 1]$ applied at each scale to mitigate distribution shift. At each resolution, two stages are used:

- **Fade-in phase:** new convolutional and upsampling blocks are smoothly integrated using η.
- **Stabilization phase:** once $\eta \to 1$, training continues until loss stabilization.

Automatic Mixed Precision (AMP) training accelerates computation and reduces memory overhead. Additionally, an **Exponential Moving Average (EMA)** of generator parameters θ_G is maintained for stability and improved inference consistency:

$$\theta^{EMA} \leftarrow \beta\theta^{EMA} + (1-\beta)\theta_G, \tag{14}$$

with $\beta = 0.999$.

D. Mini-batch and Data Sampling

Due to the high dimensionality of 3D MRI data, a mini-batch size of 1 is used. Each training step samples a paired baseline and follow-up volume (x_b, x_f) with independent latent codes (z_b, z_f). Training continues until critic loss stabilization, gradient norm consistency, and visual fidelity are achieved across anatomical planes (axial, sagittal, coronal).

E. Final Optimization Objective

The full min–max optimization is expressed as:

$$\min_{G_b, G_f} \max_{\mathcal{D}_b, \mathcal{D}_f} \left(\mathcal{L}_{D_b} + \mathcal{L}_{D_f} + \mathcal{L}_{G_b} + \mathcal{L}_{G_f}^{total}\right), \tag{15}$$

and optimized using the Adam optimizer with a learning rate $\gamma = 1e-4$, and momentum parameters $(\beta_1, \beta_2) = (0.0, 0.99)$, following WGAN-GP recommendations [27].

VI. Hardware and Runtime

Training is performed on a single **NVIDIA A100 Graphics Processing Unit (GPU)** with 40 GB memory. The full progressive training cycle requires approximately 15–20 days depending on hardware and convergence threshold.

VII. Results

This section presents preliminary qualitative observations and outlines the planned quantitative evaluation pipeline for assessing the proposed TPGGAN framework. Since the model training and subsequent downstream experiments are currently ongoing, we report early training behavior and prepared evaluation methodology, with full results to be included upon completion.

Fig. 4: TPGGAN Progressive Training

Initialize $G_b, G_f, \mathcal{D}_b, \mathcal{D}_f$, EMA weights
for resolution $r = 4 \rightarrow 128$ **do**
  for phase in {fade-in, stabilize} **do**
    for iteration $= 1 \rightarrow T_r$ **do**
      for $i = 1 \rightarrow n_D$ **do** ▷ Critic updates
        Sample real (x_b, x_f) and latent (z_b, z_f)
        $\hat{x}_b = G_b(z_b)$
        $\hat{x}_f = G_f(z_f, \hat{x}_b)$
        Compute WGAN losses and gradient penalty
        Update $\mathcal{D}_b, \mathcal{D}_f$
      end for
      Generate $(\hat{x}_b, \hat{x}_f)$
      Compute generator adversarial + L1 + SSIM losses
      Update G_b, G_f
      Update EMA weights
    end for
  end for
end for

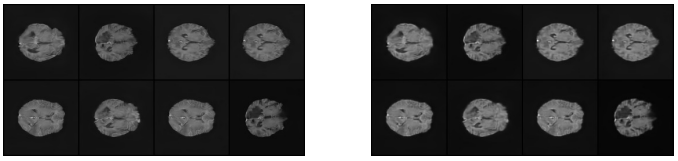


Fig. 5: Representative synthetic baseline and follow-up volume slices generated by TPGGAN at 128^3 resolution stage.

A. Preliminary Training Observations

The proposed dual-generator, dual-discriminator PGGAN architecture demonstrated stable convergence behavior under progressive resolution growth, with smooth fade-in transitions between scales and no evidence of mode collapse in early training stages [19], [25]. Generator and discriminator losses exhibited stable WGAN dynamics with gradient-penalty regularization, and EMA parameters showed reduced oscillation, indicative of training stability.

Qualitatively, generated MRI volumes exhibit increasingly coherent anatomical structure with training progression, with visible delineation of brain tissue boundaries and realistic intensity distribution. Follow-up volumes show initial evidence of temporal consistency with their corresponding synthetic baselines.

B. Planned Quantitative Evaluation

Upon full training, we will compute volumetric and perceptual similarity metrics including 3D-SSIM [26], **Peak Signal-to-Noise Ratio (PSNR)**, and **Learned Perceptual Image Patch Similarity (LPIPS)**-3D [28] between generated and real patient scans. Temporal consistency will be quantified using longitudinal SSIM and tumor progression coherence indices.

TABLE V: Planned downstream evaluation metrics

Task	Metrics
Segmentation	Dice, F1, Precision, Recall
Classification	Accuracy, Sensitivity, Specificity, AUC

C. Downstream Validation via Segmentation and Classification

To assess clinical utility, the synthesized paired MRI volumes will be used to augment real training data for two downstream tasks:

- Multi-class brain tumor segmentation
- Patient outcome classification using image + segmentation mask fusion

Segmentation accuracy will be evaluated using Dice score, F1-score, precision, and recall. Classification performance will be assessed via accuracy, sensitivity, specificity, and **Area Under the Curve (AUC)**. For segmentation baselines we will compare to established volumetric segmentation networks such as U-Net [29], V-Net [30], and nnU-Net [31].

VIII. Discussion and Limitations

The proposed TPGGAN framework introduces a novel approach for synthesizing longitudinal MRI pairs that preserve subject identity and reflect plausible tumor evolution. By combining progressive 3D adversarial training with temporal consistency constraints, the model moves beyond conventional single-timepoint GANs and provides a principled foundation for data augmentation in longitudinal neuro-oncology settings. Preliminary training behavior suggests stable optimization and promising anatomical structure emergence, supporting the feasibility of the proposed formulation.

A. Anticipated Advantages

The framework is expected to benefit downstream clinical learning tasks in three key ways:

1) **Paired augmentation**: Enables expansion of rare longitudinal tumor datasets, addressing sample scarcity.
2) **Temporal realism**: Models disease evolution rather than isolated timepoints, improving clinical relevance.
3) **Generalizable pipeline**: Compatible with segmentation and classification architectures for integrative evaluation.

These properties position TPGGAN as a generative foundation model tailored for progression-aware medical imaging studies.

B. Limitations

Despite encouraging initial behavior, several limitations warrant acknowledgment:

- **Training complexity:** Progressive 3D GANs are computationally expensive and require careful stabilization [19].
- **Limited dataset size:** The underlying BraTS longitudinal cohort remains relatively small, which may restrict generalization.

- **Lack of real clinical validation at this stage:** While the evaluation protocol includes expert review and downstream experiments, full validation will be reported upon completion of model training.
- **Potential anatomical drift:** Without explicit structural constraints on all brain subregions, subtle misalignments may occur; future iterations may incorporate deformation-aware losses.

C. Future Work

Planned extensions include:

- Full integration of synthetic pairs for tumor segmentation and classification pipelines.
- Ablation studies evaluating the role of temporal losses, EMA, and progressive scaling.
- Incorporation of 3D attention modules [32] and diffusion-based refinement [33] for sharper tumor boundaries.
- Radiologist-in-the-loop scoring and integration of clinical endpoints (e.g., survival time, treatment response status).

IX. Conclusion

This work introduces TPGGAN, a progressive 3D conditional GAN designed to synthesize anatomically consistent paired baseline and follow-up MRI scans for brain tumor progression modeling. Unlike conventional generative models that operate on single timepoints, the proposed framework explicitly learns longitudinal evolution through dual generators, cross-resolution feature fusion, and temporal consistency regularization. Early results indicate stable volumetric generation and emerging temporal coherence, suggesting that paired synthetic augmentation is a viable strategy to support research in tumor growth prediction and treatment response analysis [3].

The proposed pipeline is currently being integrated into a full downstream evaluation workflow, including tumor segmentation and multi-class outcome classification using both real and augmented paired MRI data. Comprehensive quantitative analysis, radiological review, and ablation studies will be included upon completion of ongoing experiments. As medical datasets with true longitudinal ground truth remain limited, generative augmentation of temporal imaging trajectories represents an important direction for enhancing model robustness and improving learning in neuro-oncology.

We anticipate that frameworks such as TPGGAN will play a central role in the development of clinically informed synthetic imaging models capable of supporting progression-aware computational neuro-oncology research.

Acknowledgment

The authors would like to thank the organizers of the BraTS **Longitudinal Tumor Segmentation (LTS)** challenges 2025 for providing access to longitudinal brain tumor imaging datasets that made this research possible. We also acknowledge the **Centre of Excellence (CoE) in Artificial Intelligence (AI), Netaji Subhas University of Technology (NSUT), New Delhi**, for their invaluable support throughout the course of this research.

We thank the open-source community for maintaining critical deep learning libraries used in this work, including PyTorch [34], **MONAI (Medical Open Network for AI)** [35], TorchIO [20], and associated medical imaging toolkits. Computational experiments were supported in part by academic research computing facilities.

References

[1] P. Y. Wen, D. R. Macdonald, D. A. Reardon, T. F. Cloughesy, E. A. Chiocca, M. J. B. Taphoorn, L. B. Nabors, J. G. Cairncross, J. C. Tonn, M. A. Vogelbaum, E. Galanis, J. C. Reijneveld, S. M. Chang, A. B. Lassman, M. C. Chamberlain, W. P. Mason, and M. Prados, "Updated response assessment criteria for high-grade gliomas: Response Assessment in Neuro-Oncology (RANO) working group," *Journal of Clinical Oncology*, vol. 28, no. 11, pp. 1963–1972, 2010.

[2] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 27, 2014.

[3] H.-C. Shin, N. A. Tenenholtz, J. K. Rogers, C. G. Schwarz, M. L. Senjem, J. L. Gunter, K. P. Andriole, and M. Michalski, "Medical image synthesis for data augmentation and anonymization using generative adversarial networks," in *Simulation and Synthesis in Medical Imaging (SASHIMI @ MICCAI)*, 2018, pp. 1–11.

[4] M. Frid-Adar, I. Diamant, E. Klang, M. Amitai, J. Goldberger, and H. Greenspan, "GAN-based synthetic medical image augmentation for increased CNN performance in liver lesion classification," *Neurocomputing*, vol. 321, pp. 321–331, 2018.

[5] A. Radford, L. Metz, and S. Chintala, "Unsupervised representation learning with deep convolutional generative adversarial networks," *arXiv preprint arXiv:1511.06434*, 2015.

[6] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 5967–5976.

[7] Y. Kirchhoff, P. Vollmuth, M. Foltyn-Dumitru, T. Wald, B. Kovacs, M. Zenk, K. Maier-Hein, J. Kleesiek, J. Egger, Y. Suter, M. Reyes, A. Ferreira, S. Bakas, R. Y. Huang, and J. Villanueva-Meyer, "BraTS Brain Tumor Progression Subchallenge (BraTSPRO) — 2025 BraTS Brain Tumor Progression Challenge," MICCAI BraTS Lighthouse / BraTS Challenge materials, 2025, predicting Tumor Response During Therapy. Final datasets released June 26, 2025. Accessed: June 27, 2025. [Online]. Available: https://www.synapse.org/Synapse:syn64153130/wiki/631459

[8] M. B. O. Committee and M. Lighthouse Chairs, "MICCAI 2025 BraTS & BraTSPRO Challenge Materials," Medical Image Computing and Computer-Assisted Intervention (MICCAI) challenge and workshop webpages, 2025, BraTS 2025 challenge — Predicting Tumor Response During Therapy. Accessed: June 27, 2025. [Online]. Available: https://www.miccai.org/

[9] M. S. Islam, M. A. Rahman *et al.*, "A comprehensive survey on GANs for medical image analysis," *Artificial Intelligence in Medicine*, vol. 152, p. 102637, 2024.

[10] I. Koshino, T. Fukunaga, M. Nishio, and Y. Nakajima, "Generative adversarial networks in medical imaging: A review," *Medical Image Analysis*, vol. 74, p. 102237, 2021.

[11] S. Hussain, T. Rahman, and F. Ahmed, "Longitudinal generative models for medical imaging: A review of challenges and future directions," *IEEE Access*, vol. 13, pp. 54 321–54 345, 2025.

[12] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2223–2232.

[13] J. Wu, C. Zhang, T. Xue, B. Freeman, and J. Tenenbaum, "Learning a probabilistic latent space of 3D shapes from voxel grids," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 29, 2016, pp. 82–90.

[14] A. B. L. Larsen, S. K. Sønderby, H. Larochelle, and O. Winther, "Autoencoding beyond pixels using a learned similarity metric," in *International Conference on Machine Learning (ICML)*, 2016, pp. 1558–1566.

[15] X. Yi, E. Walia, and P. Babyn, "Generative adversarial network in medical imaging: A review," *Medical Image Analysis*, vol. 58, p. 101552, 2019.

[16] X. Huang, M.-Y. Liu, S. Belongie, and J. Kautz, "Multimodal unsupervised image-to-image translation," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 172–189.

[17] A. Shrivastava, T. Pfister, O. Tuzel, J. Susskind, W. Wang, and R. Webb, "Learning from simulated and unsupervised images through adversarial training," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2107–2116.

[18] C. Han, L. Rundo, R. Araki, Y. Nagano, Y. Furukawa, G. Mauri, H. Nakayama, and H. Hayashi, "Combining noise-to-image and image-to-image GANs: Brain MR image augmentation for tumor detection," *IEEE Access*, vol. 7, pp. 156 966–156 977, 2019.

[19] T. Karras, T. Aila, S. Laine, and J. Lehtinen, "Progressive growing of GANs for improved quality, stability, and variation," in *International Conference on Learning Representations (ICLR)*, 2018.

[20] F. Pérez-García, R. Étienne *et al.*, "TorchIO: a Python library for efficient loading, preprocessing, augmenting and patch-based sampling of medical images in deep learning," *Computer Methods and Programs in Biomedicine*, vol. 208, p. 106236, 2021.

[21] R. W. Cox and J. S. Hyde, "NIfTI-1.1: A standard for neuroimaging data," *NeuroImage*, 2004, NIfTI format specification.

[22] N. Otsu, "A threshold selection method from gray-level histograms," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 9, no. 1, pp. 62–66, 1979.

[23] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "HD-BET: a robust brain extraction tool," *Scientific Reports*, vol. 9, no. 1, p. 11394, 2019.

[24] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein generative adversarial networks," in *International Conference on Machine Learning (ICML)*, 2017.

[25] I. Gulrajani, F. Ahmed, M. Arjovsky, V. Dumoulin, and A. Courville, "Improved training of Wasserstein GANs," *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, pp. 5767–5777, 2017.

[26] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Transactions on Image Processing*, vol. 13, no. 4, pp. 600–612, 2004.

[27] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *International Conference on Learning Representations (ICLR) - Poster*, 2015, arXiv:1412.6980.

[28] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, "The unreasonable effectiveness of deep features as a perceptual metric," in *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 586–595.

[29] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, ser. LNCS, vol. 9351, 2015, pp. 234–241.

[30] F. Milletari, N. Navab, and S.-A. Ahmadi, "V-Net: Fully convolutional neural networks for volumetric medical image segmentation," *International Conference on 3D Vision (3DV)*, pp. 565–571, 2016.

[31] F. Isensee, P. F. Jaeger, S. A. Kohl, J. R. Petersen, and et al., "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," *Nature Methods*, vol. 18, pp. 203–211, 2021.

[32] O. Oktay, J. Schlemper, L. L. Folgoc, M. Lee, M. Heinrich, K. V. Rens, B. Glocker, and D. Rueckert, "Attention U-Net: Learning where to look for the pancreas," in *Medical Image Computing and Computer-Assisted Intervention (MICCAI) Workshop on Interpretability of ML in Medical Imaging*, 2018, arXiv:1804.03999.

[33] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 6840–6851.

[34] A. Paszke, S. Gross, F. Massa, A. Lerer, and et al., "PyTorch: An imperative style, high-performance deep learning library," *Advances in Neural Information Processing Systems (NeurIPS) Workshops*, 2019.

[35] A. Diaz-Pinto, J. Cardoso, and et al., "MONAI: an open-source framework for deep learning in healthcare," *Nature Methods*, vol. 19, pp. 1322–1324, 2022, full author list available online. Use 'Andres Diaz-Pinto and Jorge Cardoso and MONAI Consortium' for brevity if needed.